

Insup Lee

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Research Interests

My research lies at the intersection of security and AI systems. I currently work on accelerating robot demonstrations for faster task execution with VLA policies, and aim to improve the safety and trustworthiness of physical AI.

Education

Korea University

Ph.D. in Cybersecurity

Seoul, Korea

Sep 2019 - Aug 2026

- Dissertation: Domain-Specific Generative Models for Data Augmentation in Cybersecurity
- Advisors: Sangjin Lee and Seokhie Hong

Korea University

B.E. in Cyber Defense

Seoul, Korea

Mar 2014 - Feb 2018

Work Experience

KAIST AI

Postdoctoral Researcher, LLM Innovation Research Center (Advisor: Dongsu Han)

Seoul, Korea

Sep 2026 - Present

Ministry of National Defense

Cybersecurity Officer (Captain), Republic of Korea Army

Seoul, Korea

Jun 2018 - May 2025

- Security Instructor, Akh Unit, Abu Dhabi, UAE, Apr 2024 - Mar 2025
- Security Engineer, Cyber Operations Command, Korea, Aug 2023 - Apr 2024
- Researcher, Agency for Defense Development (ADD), Seoul, Korea, Jul 2018 - Jul 2023

Publications

(C: conference, J: journal, P: preprint, †: corresponding author)

International Publications

- C5 *Schema-Agnostic Knowledge Graph Construction via Hybrid Ontology Discovery for Cyber Threat Intelligence*
S. Kim, J. Kim, D. Kang, D. Kim, and **I. Lee**[†]
Annual Computer Security Applications Conference (**ACSAC'26**), Dec. 2026.
- C4 *Mnemonic Hack: Recovering the Master Seed from Bitcoin Hardware Wallets via Side-Channel Attacks*
J. Baek, D. Bae, G. Kim, G. Ahn, **I. Lee**, S. Park, S. Hong, and H. Kim
ACM Conference on Computer and Communications Security (**CCS'26**), Nov. 2026.
- J10 *Multi-Domain Side-Channel Analysis for Anomaly Detection in Embedded Systems*
S. Park, D. Bae, **I. Lee**, J. Kim, H. Oh, S. Hong, and H. Kim
IEEE Embedded Systems Letters, 2026.
- C3 *Exploiting Per-Core Leakage: Electromagnetic Side-Channel Monitoring of Multicore Architectures*
D. Bae, S. Park, **I. Lee**, Y. Jung, K. Lee, H. Kim, and S. Hong
ACM/IEEE Design Automation Conference (**DAC'26**), Jul. 2026.
- J9 *MuCamp2: Generating Validated Cyber Campaign Variants via Constrained LLMs for Group Attribution*
K. Kim, J. Lee, **I. Lee**, I. Han, J. Lee, S. Shim, and C. Choi
IEEE Access, Vol. 14, pp. 87329-87344, Jun. 2026.
- J8 *LeakDiT: Diffusion Transformers for Trace-Augmented Side-Channel Analysis*
I. Lee, D. Bae, S. Hong, and S. Lee
IEEE Computer Architecture Letters, Vol. 25, No. 1, pp. 5-8, Jan./Jun. 2026.

- J7 *Multi-Step LLM Pipeline for Enhancing TTP Extraction in Cyber Threat Intelligence*
H. Kim, D. Lee, **I. Lee**, S. Lee, and S. Lee
IEEE Access, Vol. 13, pp. 179696-179710, Oct. 2025.
- J6 *Enhancing Modulation Classification via Diffusion Transformers for Drone Video Signal Processing*
I. Lee[†], K. Alteneiji, and M. Alghfeli
IEEE Signal Processing Letters, Vol. 32, pp. 3325-3329, Aug. 2025.
- J5 *MuCamp: Generating Cyber Campaign Variants via TTP Synonym Replacement for Group Attribution*
I. Lee and C. Choi
IEEE Transactions on Information Forensics and Security (**TIFS**), Vol. 20, pp. 6162-6174, Jun. 2025.
- J4 *UniQGAN: Towards Improved Modulation Classification With Adversarial Robustness Using Scalable Generator Design*
I. Lee and W. Lee
IEEE Transactions on Dependable and Secure Computing (**TDSC**), Vol. 21, No. 2, pp. 732-745, Mar./Apr. 2024.
- J3 *Camp2Vec: Embedding Cyber Campaign With ATT&CK Framework for Attack Group Analysis*
I. Lee and C. Choi
ICT Express, Vol. 9, No. 6, pp. 1065-1070, Dec. 2023.
- J2 *Exploiting TTP Co-occurrence via GloVe-Based Embedding With ATT&CK Framework*
C. Shin, **I. Lee**, and C. Choi
IEEE Access, Vol. 11, pp. 100823-100831, Sep. 2023.
- J1 *BAN: Predicting APT Attack Based on Bayesian Network With MITRE ATT&CK Framework*
Y. Kim, **I. Lee**, H. Kwon, G. Lee, and J. Yoon
IEEE Access, Vol. 11, pp. 91949-94968, Aug. 2023.
- C2 *Anomaly Dataset Augmentation Using Sequence Generative Models*
S. Shin, **I. Lee**, and C. Choi
IEEE International Conference on Machine Learning and Applications (ICMLA'19), Dec. 2019.
- C1 *Opcode Sequence Amplifier Using Sequence Generative Adversarial Networks*
C. Choi, S. Shin, and **I. Lee**
International Conference on ICT Convergence (ICTC'19), Oct. 2019.

Under Review

- *MAVLink Authentication Enforcement for Drone Systems Under Voltage Fault Injection*
S. Park, W. Shin, D. Bae, H. Shin, **I. Lee**, S. Hong, and H. Kim
IEEE IoTJ, under submission.
- *MoDiT: Domain-Specific Diffusion Transformers for Automatic Modulation Classification*
I. Lee[†] and M. Alghfeli
IEEE TCCN, major revision.

Domestic Publications (Korean)

- D3 S. Park, D. Bae, **I. Lee**, H. Kim, and S. Hong, "EM-Based Anomaly Detection using a Dual-Domain Approach," in Proc. of the KIISC Winter Conference (CISC-W), Nov. 2025. (Selected as an Outstanding Paper Award)
- D2 H. Park and **I. Lee**[†], "Enhanced DDoS Detection via Traffic Volume-Based Labeling and Transfer Learning," Journal of Internet Computing and Services (JICS), Vol. 26, No. 4, pp. 1-8, Aug. 2025.
- D1 K. Kim and **I. Lee**[†], "User Behavior Embedding via TF-IDF-BVC for Web Shell Detection," Journal of The Korea Institute of Information Security & Cryptology (JKIISC), Vol. 34, No. 6, pp. 1231-1238, Dec. 2024.

Patents

- C. Choi and **I. Lee**, "Method of Augmenting Cyber Attack Campaign Data to Identify Attack Group, and Security System Performing the Same," U.S. Patent Application Number. 19/244,041.
- C. Choi, **I. Lee**, C. Shin, and S. Lee, "Information Identification Method and Electronic Apparatus Thereof," Korea Patent Application Number. 10-2024-0006106, January 15, 2024.
- C. Choi, C. Shin, S. Shin, S. Seo, and **I. Lee**, "Method for Training Attack Prediction Model and Device Therefor," U.S. Patent Application Number. 18/126,005; U.S. Patent Number. US20230308462A1, September 28, 2023.
- C. Choi, S. Shin, and **I. Lee**, "Apparatus, Method, Computer-readable Storage Medium and Computer Program for Generating Operation Code," Korea Patent Application Number. 10-2019-0141865, November 07, 2019; Korea Patent Number. 10-2246797, April 30, 2021.

Other Experience

- AI Cyber Challenge (AIXCC), DARPA and ARPA-H, USA Apr 2024 – Aug 2024
- Participated in the semifinal round as a member of Team KORIA, submitting our cyber reasoning system that leverages LLMs for automated detection and patching of software vulnerabilities
- SW Outsourcing Development, KCMVP-Certified Cryptographic Module Jun 2017 – May 2018
- Implemented a cryptographic module with 25,000 LoC in C - ARIA block cipher (modes: ECB, CBC, CTR), hash functions (SHA-256, SHA-512), and HMAC-based DRBG for Windows (.dll) and Linux (.so)

Honors and Awards

- KU Graduate School Achievement Award Feb 2026
Korea University, Seoul, Republic of Korea
- Outstanding Paper Award Nov 2025
CISC-W'25, KIISC
- Certificate of Commendation (UAE-ROK Engagement Program) Mar 2025
United Arab Emirates Ministry of Defense
- Ambassador's Commendation Mar 2025
Embassy of the Republic of Korea to the United Arab Emirates
- Full Tuition Scholarship (Korea University) Mar 2014 – Feb 2018
Ministry of National Defense, Republic of Korea

Academic Service

Journal Reviewer

- IEEE Transactions on Dependable and Secure Computing (TDSC), 2025
- IEEE Transactions on Information Forensics and Security (TIFS), 2026
- IEEE/ACM Transactions on Networking (ToN), 2026
- IEEE Transactions on Communications (TCOM), 2025, 2026
- IEEE Journal on Selected Areas in Communications (JSAC), 2025, 2026

Teaching Experience

- Lecturer, Computer Networks (SCS302, Fall 2025), Korea University Sep 2025 - Dec 2025
- Instructor, Penetration Testing - Intermediate, UAE Ministry of Defense Sep 2024
- Instructor, Penetration Testing - Basic, UAE Ministry of Defense Jul 2024

Invited Talks

- Cybersecurity and Hacking in the AI Era March 2026
Major Exploration for Freshman, Sungshin Women's University
- Exploring Career Paths in Security Research November 2025
Major Exploration for Freshman, Sejong University

Mentoring Experience

- Seonwoo Kim (Major at Ministry of National Defense) Feb 2026 – Present
Project: LLM-based knowledge graph construction for CTI [C5]
- Sujin Park (Ph.D. Student at Korea University) Jun 2025 – Present
Project: Side-channel analysis for anomaly detection [J10]
- Hyoungrok Kim (M.S. Student at Korea University) Mar 2025 – Sep 2025
Project: LLM-based TTP extraction for CTI [J7]
- Hyunjun Park (Navy Lieutenant at Ministry of National Defense) Nov 2024 – Feb 2025
Project: DDoS detection via transfer learning [D2]
- Kangmun Kim (First Lieutenant at Cyber Operations Command) Jan 2024 – Sep 2024
Project: Web shell detection via user behavior embedding [D1]

Reference

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Seokhie Hong

Vice President, SmartM2M
Former Professor, School of Cybersecurity, Korea University
Relationship: Ph.D. Advisor
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Jongin Lim

Professor Emeritus, School of Cybersecurity, Korea University
Former Special Presidential Advisor for Cybersecurity, Republic of Korea
Relationship: Undergraduate Advisor
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